

OECD/NRC BENCHMARK BASED ON NUPEC PWR SUBCHANNEL AND BUNDLE TESTS (PSBT)

Assembly Specifications and Benchmark Database Volume I Addendum

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**Incorporated Administrative Agency
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(JNES)**

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1. Location of Process Measurement

First of all, it shall be made one correction. During the PSBT-1 meeting in Pisa, incorrect information has been provided that is Pressure measurement was done at the outlet of the test vessel. As shown in Table 1 and Figure 1, Pressure measurement was done at the inlet of the test vessel not the outlet of it.

Table 1 Location of Process Measurements

Item	Location	Drawings
Pressure	The pipe between up-stream of the inlet nozzle of the test vessel and down-stream of the pre-heater. The pipe size is 100AS160. The pipe length from the measurement location to the vessel coolant inlet is not clarified.	Appendix-A
Fluid Temperature		
Flow		
Power	The current and voltage over the test section	

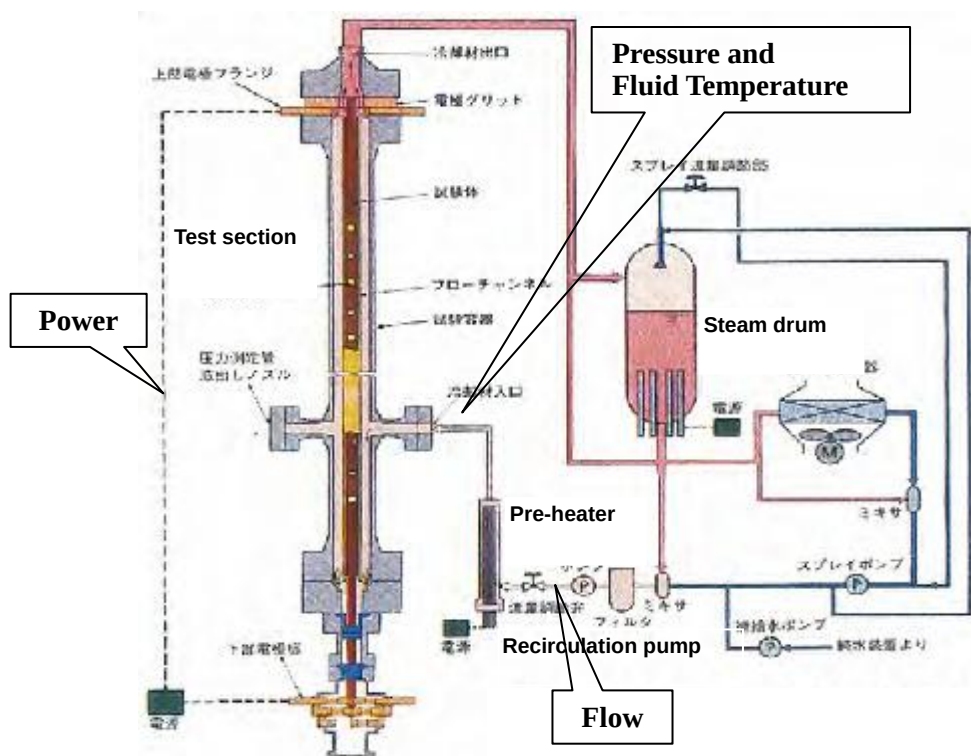


Figure 1 System Diagram of Test Facility for NUPEC PFBT

2. Test Vessel and Flow Channel Structures

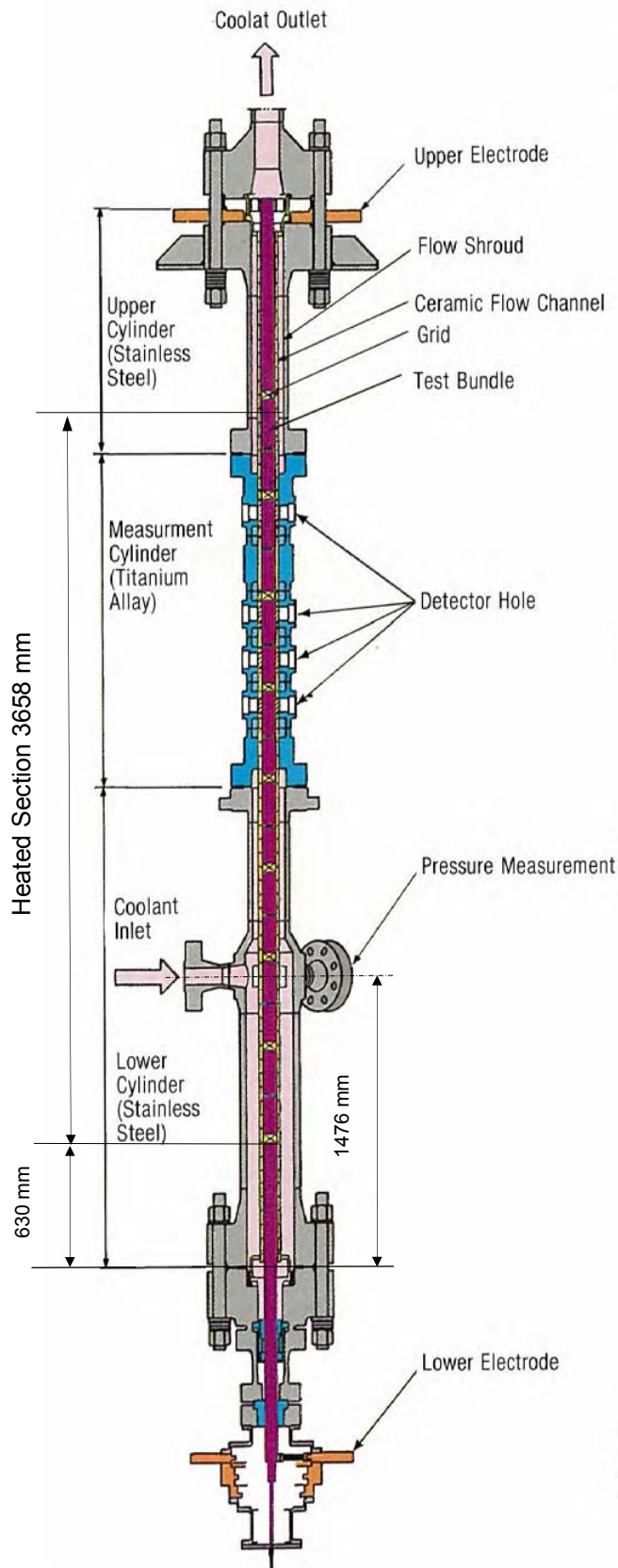


Figure 2 Test Vessel

Table 2 Test Vessel and Flow Channel Structure Data

Flow Channel	Inner length (mm)	64.9
	Thickness (mm)	5.0
	Material	Alumina
Flow Shroud	Inner length (mm)	75.1
	Thickness (mm)	10*
	Material	SUS304
Test Vessel (Upper Cylinder)	Outer diameter(mm)	216.3
	Thickness (mm)	23.0
	Material	SUS304
Test Vessel (Lower Cylinder)	Outer diameter(mm)	318.5
	Thickness (mm)	33.3
	Material	SUS304

* Measured value from drawings (Appendix-B).

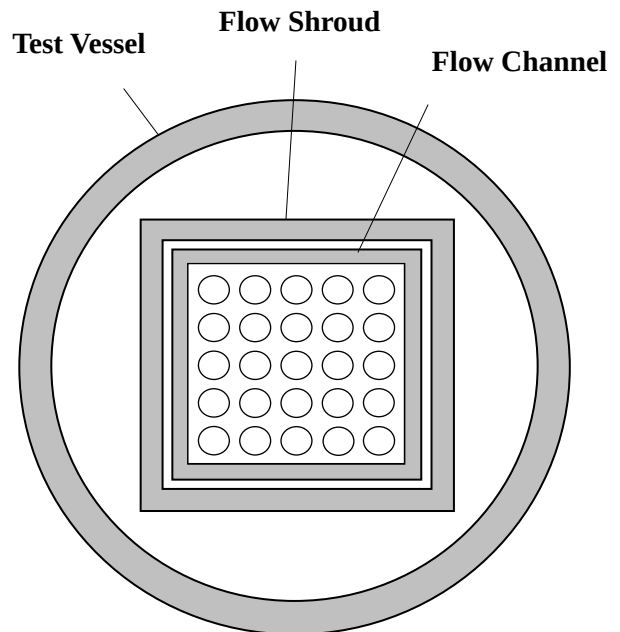


Figure 3 Cross-Section of Test Vessel

3. Suggested Analysis Condition on Transient Exercise I-3

Analysis condition on transient exercise I-3 in PSBT benchmark is suggested in Table 3.

The location of the fluid Temperature measurement is the pipe between up-stream of the inlet nozzle of the test vessel and down-stream of the pre-heater, where the pipe size is 100AS160.

Therefore, D (the pipe inner diameter) is 100 mm. But the L (the pipe length from the coolant inlet to the measurement location) is not clarified.

It is suggested that L=0 for the reference analysis condition and L as a parameter for the parameter survey analysis if a participant wishes to do it. And same condition on the inlet Flow measurement.

Table 3 Suggested Analysis Condition on Transient Exercise I-3

Item		Analysis condition	
		Reference	Parameter Survey
Location of Measurement	☐ Pressure	L=0, D=100mm	N/A
	☐ Fluid Temperature	L=0, D=100mm	L as a parameter
	☐ Flow	L=0, D=100mm	L as a parameter
	☐ Power	Test Bundle	N/A
Geometry of Coolant Inlet	☐ Elevation of the inlet nozzle (referred to the bottom of heated section)	846 mm	
	☐ Height of the downcomer	1476 mm	
	☐ Height of the lower unheated part of the bundle	630 mm*	
	☐ Outer diameter of the vessel in this region	318.5mm	
	☐ Thickness of the vessel in this region	33.3mm	
	☐ External square side of the flow shroud	95.1mm	
Geometry of the lower part of the circuit where the flow reverses from downwards to upwards			
- Height		100 mm* (to be assumed no obstacle)	
- Elevation of the Center		Bottom of the downcomer	

(Note) L: Length of the pipe from the coolant inlet nozzle to the measurement location,
D: Inner diameter of the pipe.

* Measured value from drawings (Appendix-B)

4. Error Source of Void Measurements

Error source of void measurements for chordal averaged and CT averaged is summarized in Table 4.

The correlations between the chordal averaged value and the channel averaged value given by the best fit curves as shown in Figure 4 were introduced respectively for the high pressure conditions and the low pressure conditions. The uncertainty of this correlation is evaluated to be less than 18 kg/m^3 and this value was regarded as three standard deviations (3σ). Therefore one standard deviation (1σ) of it is 6 kg/m^3 .

Void measurements uncertainties provided in Table 4 correspond to one standard deviation (1σ).

It is suggested to use a two standard deviations (2σ) uncertainty bound for the 95% confidence in the code-to-data comparisons for the single-measurement tests and a three standard deviations ($3\sigma = \sqrt{(2\sigma)^2 + (2\sigma)^2}$) uncertainty bound for the duplicated-measurement tests.

Table 4 Error Source of Void Measurement

Error source		Chordal Averaged		CT Averaged
		Steady-state	Transient	
γ-ray Measurement	Effect of surrounding condition (magnetic-field and temperature) on measurement system	0.1%	0.1%	0.1%
	Randomness of γ-ray source decay	0.02%	0.2%	0.1%
	Correction error due to back ground	0.0%	0.0%	0.0%
	Correction error due to counting loss	<0.5%	<0.5%	<0.1%
	Calibration error	0.1%	0.1%	0.1%
	Correction error due to attenuation by surrounding water	0.0%	0.0%	-
	Correction error due to scattering from multi γ-rays	<0.2%	<0.2%	-
Total		<0.55%	<0.6%	<0.2%
Sub-channel Density	Transfer to density	<9 kg/m ³	<10 kg/m ³	<15 kg/m ³
	Distribution error to Sub-channel	<5 kg/m ³	<5 kg/m ³	-
Correlation error from Chordal averaged to CT averaged		<6 kg/m ³	<6 kg/m ³	-
Sub-channel Density		<20 kg/m ³	<21 kg/m ³	<15 kg/m ³
Sub-channel Void*		0.040	0.042	0.030
Uncertainty (1σ)		4 %	5 %	3 %

* Reference averaged density is 500 kg/m^3 (see Figure 4).

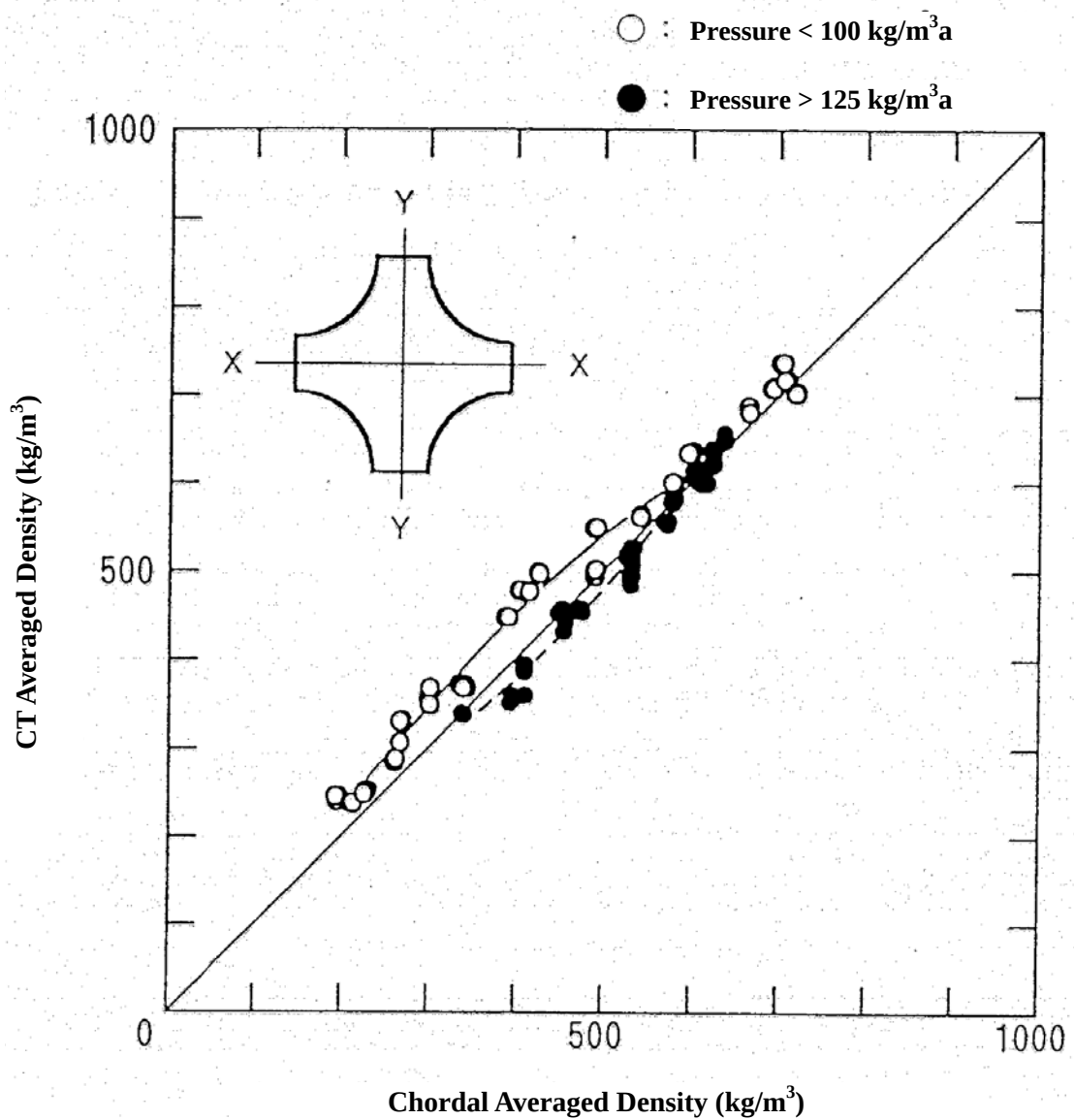


Figure 4 Relation between CT Averaged Density and Chordal Averaged Density in S1

5. Water Flow in Thimble Rods

Thimble rods are filled with stratified water.

Actually, this flow condition is made by with the tube structure open in the bottom and a pin hole in the top. Therefore, the same structure may be applied in PSBT but it couldn't be confirmed because no drawings on thimble rods in the NUPEC Test Report.

Appendix

Appendix-A Measurement diagram of the test loop of PSBT

Appendix-B Drawings of the test vessel of PSBT